13.3. A Simple Supersymmetry Model
Let

$$H = \omega \left(a^\dagger a + b^\dagger b \right)$$

where a are b are have the algebra of a single boson and fermion mode, respectively:

$$\left[a, a^\dagger \right] = 1 \quad \text{and} \quad \left\{ b, b^\dagger \right\} = 1$$

a and $a^\dagger$ are independent of b and $b^\dagger$, so they commute:

$$\left[a, b \right] = \left[a, b^\dagger \right] = 0$$

The eigenstates of this Hamiltonian are

$$|n_B, n_F\rangle = \frac{\left(a^\dagger\right)^{n_B}}{\sqrt{n_B!}} \left(b^\dagger\right)^{n_F} |0,0\rangle$$

n_B can be any nonnegative integer, while n_F is 0 or 1. The state has n_F "fermions" and n_B "bosons." These are not really bosons and fermions of course. To do that, a and b would have to be functions of the momentum $\mathbf{p}$. Rather, they have the properties of fermions and bosons with a single momentum.

The state with one fermion has the same energy as the state with one boson. When there is extra degeneracy, you know there must be extra symmetry.

(a) Define the operator

$$Q = b^\dagger a$$

Show that Q and $Q^\dagger$ commute with the Hamiltonian, and that

$$\left\{ Q, Q^\dagger \right\} = \beta H$$

What is the constant β? Set

$$Q = Q_1 + iQ_2 \quad \text{and} \quad Q^\dagger = Q_1 - iQ_2$$

Show that

$$\{Q_i, Q_j\}$$

is proportional to $\delta_{ij} H$.

(b) $|1,0\rangle$ is the one-boson state. Show that $Q|1,0\rangle$ is proportional to $|0,1\rangle$.

> Note: These two states have the same energy, and Q changes one into the another. A symmetry like this that changes a boson state into a fermion state is called a **supersymmetry**.

(c) On the usual vector space of one-dimensional wave functions, or equivalently the space for which the states $|x\rangle$ are a basis, you can define a and $a^\dagger$ in the usual way, but there is no way to define operators with the properties of b and $b^\dagger$. Instead, you have to add a discrete quantum number, so that a basis is $|x, m\rangle$, where $m = \pm\frac{1}{2}$. Think of the states as a column vector, or spinor, with two rows:

$$\langle x, m | \psi\rangle = \psi_m(x)$$

and

$$\psi(x) = \begin{pmatrix} \psi_{+\frac{1}{2}}(x) \\ \psi_{-\frac{1}{2}}(x) \end{pmatrix} = \psi_{+\frac{1}{2}}(x)\chi_+ + \psi_{-\frac{1}{2}}(x)\chi_-$$

with

$$\chi_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \chi_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Set

$$a = \frac{1}{\sqrt{2m\omega}}(m\omega x + ip) \qquad \text{and} \qquad b = \frac{1}{2}(\sigma_x - i\sigma_y)$$

Show that b and $b^\dagger$ satisfy the correct anticommutation rule, and then write the Hamiltonian in terms of x, p, and the Pauli matrices.

(d) What is the two-component wave function for the lowest energy state, and what is its energy?

13.4. Supersymmetry Model with Interaction

Problem 13.3 describes a boson excitation and a fermion excitation with the same energy (level spacing). But those were just independent free particles. Here is a more interesting model (but still only a single mode). Again consider a space of wave functions that are two component spinors. Let[12]

$$Q_1 = \frac{1}{\sqrt{8m\omega}}\left(p\sigma_y - m\omega^3 x^3 \sigma_x\right) \qquad Q_2 = \frac{1}{\sqrt{8m\omega}}\left(p\sigma_x + m\omega^3 x^3 \sigma_y\right)$$

(a) Show that

$$\left\{Q_i, Q_j\right\} = \beta\delta_{ij} H$$

where

$$H = \frac{p^2}{2m} + V_1(x) + V_2(x)\sigma_z$$

What is H? What is the constant β?

(b) It follows that Q_1 and Q_2 are symmetries of this problem: $[Q_i, H] = 0$. Why? Show that there is an eigenstate of H with energy zero. What is its wave function?

Hint: If $H|\psi\rangle = 0$, then $Q_i|\psi\rangle = 0$ also. Why?

(c) The state with zero energy is the ground state. Why?

13.5. Hamiltonian for Boson Fields

Derive the form (13.122) of the Hamiltonian for a free Bose-Einstein field in terms of the creation and annihilation operators.

13.6. Fermion Pair Correlations

Consider a collection of noninteracting nonrelativistic spin-half fermions, confined to a cubic box with side L, and satisfying periodic boundary conditions at the walls, as in Section 13.2.2. The particle annihilation operators are $b_s(\mathbf{p_n})$, where s is the spin index and the components of the momentum are $p_i = 2\pi n_i/L$

[12] Suggested in a slightly different form by J. M. Cornwall.